

Artificial Intelligence in Orthopedic Surgery: Current Applications, Challenges, and Future Directions

Introduction

AI in orthopaedics serves as a complementary tool, emphasizing how every aspect of AI can help clinicians to assess the patients in all clinical domains, rather than replacing them. This review focuses on using AI for orthopedics in planning, intraoperative robotics, smart implants, and bone regeneration. The core of the study focuses on the role of AI in orthopedic surgeries, including its advantages and disadvantages, as well as the concerns surrounding its risks and future applications in orthopedic practice. The primary goal is to summarize available information about what AI is, where it is currently applied in orthopaedics, what benefits it provides, and what hurdles have to be overcome prior to implementing it in routine practice. AI has radically transformed orthopedic surgery, through improved diagnostics, surgical planning, intraoperative assistance and post surgical rehabilitation.

1. AI in Diagnostic Imaging and Fracture Detection

Machine learning has shown a great outcome in orthopaedics. It is used to automatically detect the fractures and in classifications on radiographs. Deep learning has substantially supported clinicians using radiological imaging in orthopaedic trauma, arthroscopic assessment and long-term arthroplasty surveillance. DL supports detection of fractures on X-rays and injuries like spinal stenosis and segmentation of meniscal and cartilage tissue on MRI.

Neural networks like FracNet and Convolutional Neural Networks like AlexNet and YOLOv8-AM demonstrated a high level of accuracy equivalent to orthopaedic surgeons. They are used in detection of fractures on radiographs based on limited datasets, especially distal radius fractures. YOLOv8-AM is a major improvement in fracture detection in children. One of the AI models performs better than manual techniques, attaining over 98% accuracy, high Dice coefficients, and low surface errors, even in complex fractures.

Machine learning algorithms show impressive results at such tasks as image segmentation, localization and classification. They were successfully used in cartilage and meniscus segmentation, spine and vertebrae localization, fracture and foot type classification. Algorithms

demonstrate high accuracy in detecting such conditions as developmental dysplasia of the hip, meniscal tear, ACL tear, and osteoarthritis, with some performing comparably to specialized physicians under laboratory conditions.

AI can differentiate between benign tumors and malignant tumors by analysis of multiple radiological images. Beyond imaging, ML models now assess tumor pathology at a cellular level, offering more accurate prognoses than conventional methods. NLP and LLM are a major development in research and education, as they extract and structure information from large datasets. In orthopaedics, they are used in collecting operative approaches, fixation method and postoperative important notes. In addition, they analyse radiology reports to identify fracture injuries.

2. AI in Preoperative Planning and Implant Selection

AI can generate 3D anatomical models from CT and MRI scans, reducing manual segmentation time and improving preoperative planning. Through orthopedic operations AI can reduce time to reconstruct 3D bone image from 30 minutes in the traditional way to about 233 seconds.

AI-powered platforms such as Enhatch provide surgeons with real-time, patient-specific models to optimize implant alignment and predict surgical outcomes. Stryker's Mako system uses CT-based 3D modeling to optimize implant size and alignment, reducing intraoperative errors and improving long-term outcomes in joint replacement surgeries. Several surgeons have achieved submillimeter precision in implant positioning using real-time imaging combined with preoperative models, especially in spinal and pelvic surgeries.

AI models can achieve femoral and tibial implant size prediction accuracy rates of 82.2% and 85.0%, respectively, compared to conventional manufacturer default plans showing 68.4% and 73.1% accuracy. ML models demonstrate the capability to reduce surgeon corrections to preoperative plans by 39.71% compared to manufacturer default plans, with improvements reaching 47.95%. Using VR with AR technologies, helping surgeon to minimize complications, save time, lowering costs and optimization of surgical path which are benefits in complex operations.

3. AI in Intraoperative Guidance and Robotic Systems

AI robotics can assist in orthopedic surgery like total knee arthroplasty and spinal surgeries. AI-powered robotic systems achieve sub-millimeter accuracy in implant positioning and bone cuts, resulting in reduced soft-tissue damage, and fewer surgical complications. Clinical studies report a reduction of up to 30% in operative time, 35% less blood loss. AI-assisted pedicle screw placement in spinal surgery achieves accuracy rates approaching 97%.

Deep learning architectures can achieve sub-100 ms inference times while maintaining high segmentation accuracy. Real-time ultrasound imaging, combined with CNNs and GANs, enables efficient segmentation of bone surfaces and bone shadows. These methods achieve high accuracy above 95% and localization errors as low as 0.11-0.2 mm. Deep learning-based acoustic sensing can detect critical events like drill breakthrough in milliseconds, preventing injury to vital structures and outperforming human reaction times.

Zimmer Biomet's Persona IQ knee implant monitors gait patterns, step count, and load distribution to transmit data to clinicians, which allows them to identify complications such as loosening, infection, or abnormal biomechanics before clinical symptoms arise. Smart implants and load-sensing devices provide continuous, objective feedback on structural integrity and biomechanics postoperatively, enabling tailored rehabilitation and early complication detection.

4. AI in Postoperative Care and Rehabilitation

ML can predict a patient's length of stay in hospital and the time needed for recovery. ML can analyse large datasets, so it can predict the fracture risk and the postoperative outcomes after injuries like knee arthroplasty. AI assist in postoperative care by detecting and reducing complications early and improving prediction accuracy.

With the development of AI it has become promising in rehabilitation by providing real time exercise and identifying responses and mitigating functional decline. This is achieved by combining gait analysis, imaging techniques, and VR. Gait analysis is considered one of the most important tools for assessing rehabilitation. With the help of AI, remote monitoring, exercise administration, and condition evaluation are now possible without geographical limitations.

Using LLMs like ChatGPT in discharge documents reduced administrative workload on physicians and accelerated the process. GAI appeared to write discharge documents faster than clinicians while maintaining high quality.

5. Shared Limitations and Challenges

AI diagnosis depends on the data that we train them on. So, if we trained them on adult fracture radiographs, AI will diagnose child fracture wrongly. AI training by data from patients in some places so if AI work in another patient in another places anatomical differences will making their outputs less effective for the latter patients. The majority of studies are retrospective, and there is a lack of prospective studies. Most existing studies are single-centre, meaning that their findings are not representative of the wider patient population.

The most significant challenge in using AI is the black box which is the difficulty in understanding how algorithms make their decisions, which lead to poor interpretability, thereby limiting their clinical trustworthiness. Black box means that AI provides diagnosis without explaining the reasoning behind them. Doctors must supervise AI systems to avoid wrong diagnosis. The concept of informed consent becomes increasingly complicated when AI systems act like black boxes, so that neither the patient nor the physician thoroughly comprehends the underlying decision-making process.

Sensitive patient data are collected and stored by AI, and that is a huge risk. Because advanced technologies can reach these data even if it is anonymized. Clinical testing often occurs within controlled environments, hindering the validation of devices in real clinical practice with variable clinical scenarios. The residents trained solely by robotic-assisted systems performed TKA many times less accurately than residents trained by open techniques. For these reasons, it is important that trainees also practice conventional methods even while building their expertise in AI-assisted surgery. If the artificial intelligence technical failures or system shutdowns in the operating room, this represents a further risk requiring attention.

6. Future Directions

AI must be tested in real clinical settings, not just in theoretical studies. We have to test the AI system outside of a controlled laboratory, necessitating standardized regulatory guidelines to validate AI systems prior to implementation in clinical practice. Future research should focus on large-scale clinical trials to validate AI technologies and their impact on patient outcomes.

Future research is focusing on improving data quality through multicenter studies, developing explainable and transparent AI models, integrating multimodal data, and combining AI with robotics systems for personalized care. FL is a promising development in spine surgery. It is a privacy preserving technology and it connects all healthcare facilities by sharing model information rather than patient data. AI systems should be carefully monitored and regularly updated to reduce bias and better represent all patients.

AI in orthopedic surgery is moving toward more independent and integrated systems with a focus on autonomous robots, multi-pattern data integration, and global collaboration among medical institutions.

Conclusion

AI cannot replace doctors, but they can work more efficiently by using it. It serves as a complementary tool for doctors in diagnosis, prediction of injuries, and administrative work. AI is already beginning to transform orthopaedic practice and has significant potential to improve diagnosis, surgical outcomes, and personalized care. However, it is not ready to replace clinical expertise, as important technical, ethical and validation challenges remain. To guarantee safe and successful adoption, these issues must be resolved through standardized development frameworks, strong multicenter partnerships, and clear regulations.

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